

Module 4

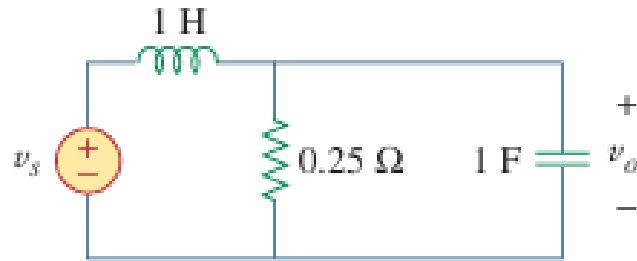
Circuit Analysis in the S Domain

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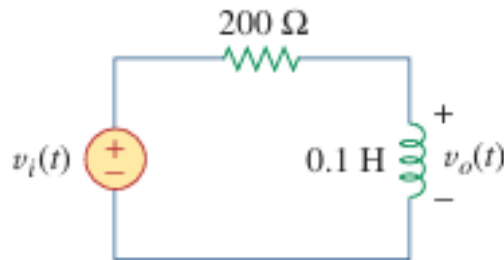
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Tutorial

1. Find the transfer function of the below. Show that the circuit is a lowpass filter.



2. Determine the type of filter and calculate the corner frequency f_c .



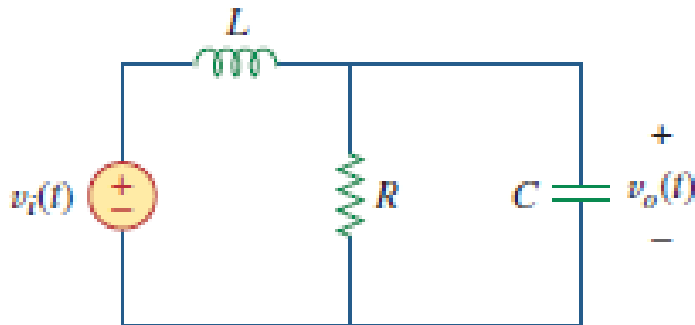
3. Determine the range of frequencies that will be passed by a series RLC bandpass filter with $R = 10\ \Omega$, $L = 25\text{ mH}$ and $C = 0.4\ \mu\text{F}$. Find the quality factor.

4. The tuner for an FM radio requires a band-pass filter with a centre frequency of 100 MHz (frequency of the FM station) and a bandwidth of 2 MHz . Design a filter circuit to accomplish this and evaluate its cut-off frequencies.

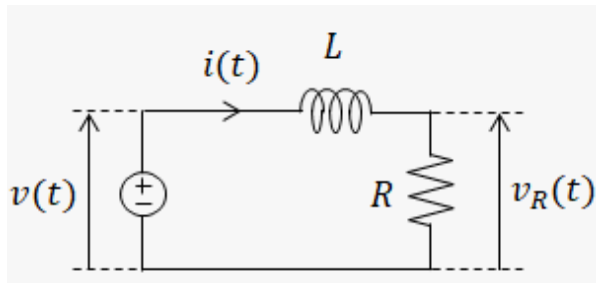
Tutorial

5. Design a bandpass filter with a lower cutoff frequency of 20.1 kHz and an upper cutoff frequency of 20.3 kHz. Take $R = 20 \text{ k}\Omega$. Calculate L , C , and Q .

6. Determine what type of filter is shown in Figure. Calculate the corner or cut-off frequency. Take $R = 2 \text{ k}\Omega$, $L = 2 \text{ H}$, and $C = 2 \text{ }\mu\text{F}$.

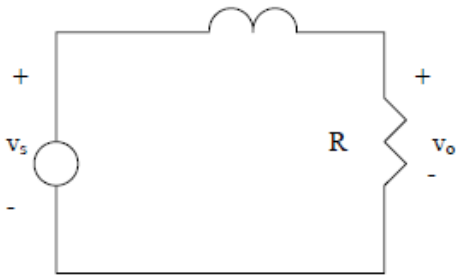


7. Show that a series LR circuit is a low pass filter if the output is taken across the resistor. Calculate the corner frequency if $L = 2 \text{ mH}$ and $R = 10 \text{ k}\Omega$

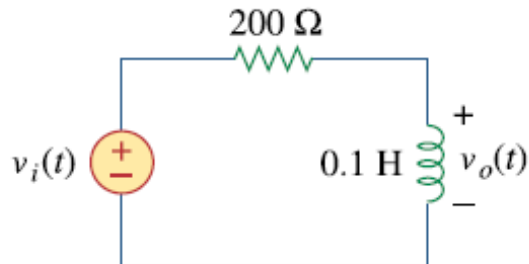


Tutorial

8. Design an RL low pass filter that uses a 40 mH coil and has a cut-off frequency of 5 kHz.



9. Determine what type of filter is in Figure. Calculate the corner frequency in Hz.



10. In a highpass RL filter with a cut-off frequency of 100 kHz, $L = 40\text{ mH}$, find R .

11. Identify the type of filter in figure. Obtain the transfer function and calculate the corner frequency if $R_1 = R_2 = 100\ \Omega$ and $L = 2\text{ mH}$.

